# Research plans and literature reviews

Lesson 8, 90 minutes. Combining tree-ring evidence

# Learning objectives

**Separate evidence for dates, environment and causes.**

Interpret agreement and disagreement among sources.

Develop a question and a plan or literature review.

# Three applications and a shared procedure

| Application | Main focus | Shared checks |
| --- | --- | --- |
| Dendroarchaeology | Objects, buildings, activity | Records, dating, comparisons, limits |
| Dendroecology | Trees, stands, disturbance | Records, dating, comparisons, limits |
| Dendroclimatology | Regional, seasonal climate | Records, dating, comparisons, limits |

R30 Historic England

R52 Yonenobu & Eckstein (2006)

# Observation, dating and interpretation

**Observation: directly measured or inspected features.**

Dating: placement by crossdating and calendar anchoring.

Interpretation: inference about cutting, disturbance, climate or activity.

# Wood and archaeological time

**Ring formation** — **Felling** — **Processing / use** — **Deposition**

Reuse and repair can separate timber and building dates.

Check the evidence connecting these stages.

Conceptual stages, not measured dates for a particular object.

# Linking climate and history

**Coinciding changes can motivate a causal hypothesis.**

Examine food, water, land use, institutions and other mediating processes.

Match region, season and temporal resolution.

R52 Yonenobu & Eckstein (2006)

# Case: wooden containers and dated wooden slips

**Maeda and colleagues (2024) compare Nara wooden containers and wooden slips.**

Tree-ring dates are compared with written dates.

Read how different sources constrain feature chronology.

R55 Maeda et al. (2024)

# Comparing outer-ring and written dates

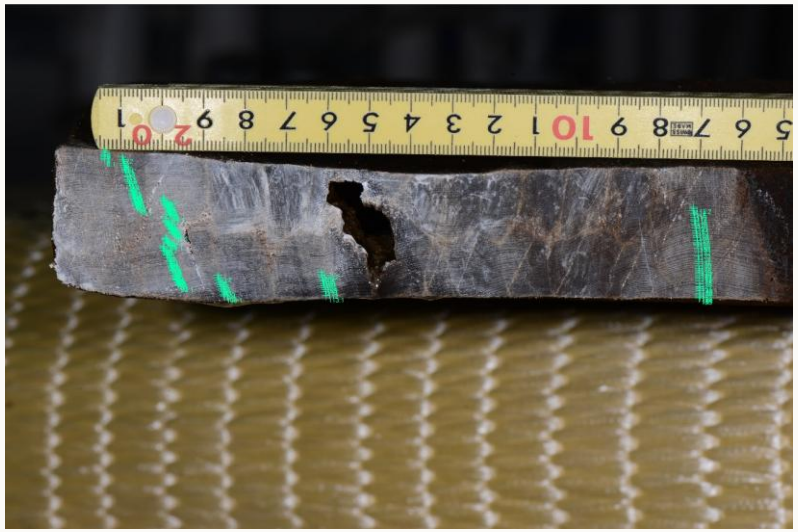
**What event does the written year describe?**

What outer wood is preserved in the ring-dated object?

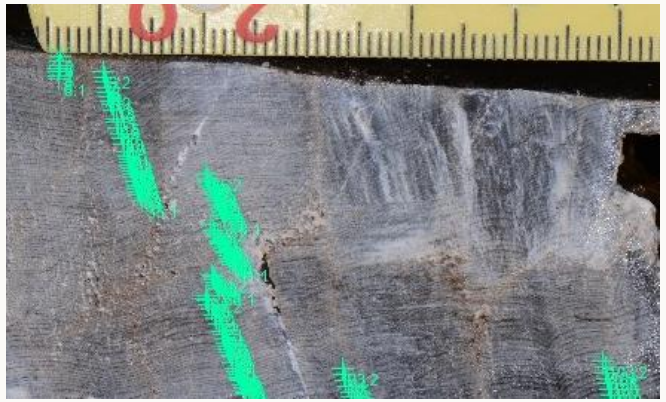
Examine association, duration of use and redeposition.

# From an excavated sample to its data

**Whole image**



**Detail cropped from left side**



## Sample 29ho018

Short marks indicate measurement positions and ring boundaries.

Use the sample ID to connect the image with the numeric record.

Check the source records for species and calendar dates.

Source: Nabunken, 29ho018

CC BY 4.0

# An ecological view of growth changes

**Suppression may involve climate, insects or competition.**

Combine scars, recruitment, mortality and management records.

Examine comparison trees and survivorship bias.

R35 Nowacki & Abrams (1997)

R45 Walker et al. (2014)

# The scope of a climate reconstruction

**State region, season, climate variable and units.**

Check calibration, verification and response stability.

Test the links from local climate to broader social change.

R52 Yonenobu & Eckstein (2006)

# Exercise A: interpret three timber dates

**All examples are fictional. A building record mentions 1210.**

| Timber | Last preserved ring | Bark | Location |
|---|---|---|---|
| A | 1187 | Absent | Part 1 |
| B | 1199 | Present | Part 2 |
| C | 1213 | Absent | Part 3 |

**8 minutes: claims, limits and additional evidence.**

Synthetic teaching data / schematic, not empirical results.

# Exercise A: the limits of the conclusion

**A: timber use after formation of the 1187 ring; felling unresolved.**

B: the 1199 ring formed; inspect season and interval before use.

C: wood formed through 1213; reassess its relation to the 1210 record.

Synthetic teaching data / schematic, not empirical results.

# Sources of uncertainty

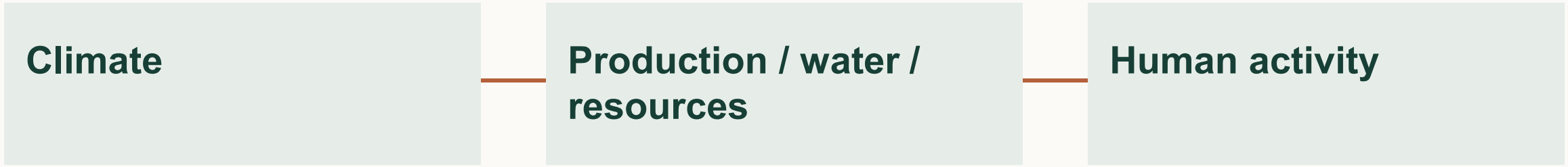
| Stage | Questions to check |
| --- | --- |
| Measurement / dating | Boundaries, units, dating evidence |
| Preservation / sampling | Missing outer wood, biased samples |
| Interpretation | Post-felling intervals, alternative causes |

R30 Historic England

R52 Yonenobu & Eckstein (2006)

# Diagramming a causal hypothesis

**Show proposed processes connecting climate, trees, production and activity.**




Consider management, land use, institutions and other pathways.

**Arrows indicate hypotheses, not proven causal effects.**

Conceptual diagram. Connections are hypotheses to test.

R52 Yonenobu & Eckstein (2006)

# Making a question specific

**Specify place, period and material.**

State competing hypotheses and predicted observations.

Identify evidence that could change the conclusion.

# Planning sources and sampling

**Check source location, preservation and access conditions.**

Plan trees, sites, comparisons and sampling positions.

Prepare alternatives if the preferred material is unavailable.

# Matching questions to methods

**Calendar dating needs crossdating and an anchored reference.**

Disturbance studies combine growth, injuries, populations and comparisons.

Climate estimation needs instrumental calibration and independent testing.

# Tracing papers, reports and datasets

**Move from the abstract to methods, materials, figures and limits.**

Record dataset version, identifiers and the subset used.

Distinguish primary reports from introductory or secondary accounts.

R55 Maeda et al. (2024)

R56 Nabunken (2024)

# Presenting figures, data and sources

**Make plots with clear axes, units and sample depth.**

Distinguish measurements, synthetic data and schematics.

Check the applicable conditions and credits for external figures.

# Exercise B: develop a plan or review

**State the question, sources, comparisons and limits of your plan or review.**

| Item | What to include |
|---|---|
| Question and target | Place, period, competing explanations |
| Sources and approach | Methods or literature comparison |
| Limits and further evidence | Alternatives, observations that change the claim |

**8 minutes: draft 5, discuss 2, revise 1.**

# Read an example against the rubric

**A last ring dated 1750 does not alone fix construction (fictional example).**

Find the evidence, repair or reuse alternatives, and limits in the short example.

Include an alternative explanation and a specific next check in your own plan.

# Final report and assessment

**Use the final 45 minutes to write, check and submit. Address one question in about four paragraphs.**

| Assessment criteria | What to show |
| --- | --- |
| Question and scope | Specify the place, period and material |
| Materials and method / comparison | Explain how to date the rings and compare alternatives |
| Evidence / conclusion and limits | Identify evidence, limits and what to check next |

**The report determines the grade. Figures are optional. There is no page or reference minimum.**

Guide: plan 5 min, write 30 min, check 10 min. Handouts and notes allowed.

# Conclusions and next checks

**State the conclusion supported by the available evidence.**

Identify remaining alternatives and uncertainty.

Name one further observation that could change the conclusion.

# Keeping records for reanalysis

**Preserve identifiers, original measurements, units and missing values.**

Track the relationship among processing, figures, text and versions.

Clarify supplied materials and release scope in collaborative work.

# Sources for continued learning

**Use bilingual terms to search papers and data.**

Connect Nabunken records, tree-ring databases and primary papers.

Continue testing explanations against observed material.

# Final reflection

**How do you distinguish a ring date from an event date?**

Can you compare two explanations for one observation?

What evidence could change your plan or review?

# Tree rings, environment and historical questions

**Record observations, verify dates and compare explanations.**

Move among specimens, data and literature.

Present a question together with the limits of its answer.